



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2023 Year 12 Course Assessment Task 4 (Trial Examination)

Wednesday, 16 August 2023

General instructions

- Working time – 3 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I - 10 marks

- Mark your answers on the answer grid provided.
- Attempt Questions 1–10
- Allow about 15 minutes for this section

SECTION II - 90 marks

- Commence each new question on a new booklet. Write on both sides of the paper.
- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class: (please ✓)

- ☐ 12MXX.1 – Mr Ho
- ☐ 12MXX.2 – Mr Sekaran
- ☐ 12MXX.3 – Ms Ham

Marker's use only

QUESTION	1-10	11	12	13	14	15	16	Total	%
MARKS	$\overline{10}$	$\overline{13}$	$\overline{13}$	$\overline{12}$	$\overline{17}$	$\overline{18}$	$\overline{17}$	$\overline{100}$	

Section I

10 marks

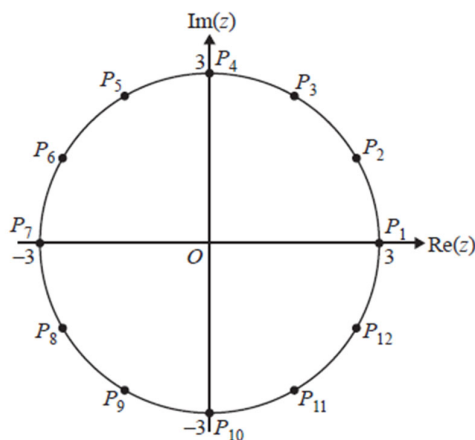
Attempt Questions 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided.

Questions	Marks
1. Which of the following expression best represents $4 + 3i$?	1
(A) $5e^{36.679i}$ (B) $25e^{0.644i}$ (C) $5e^{0.927i}$ (D) $5e^{0.644i}$	
2. The complex numbers z , iz and $z + iz$, where z is a non-zero complex number, are plotted in the Argand plane, forming the vertices of a triangle. Which of the following is the area of the triangle?	1
(A) $ z $ (C) $\frac{ z ^2}{2}$ (B) $ z + z ^2$ (D) $\frac{\sqrt{3}}{2} z ^2$	
3. Which of the following statement is true?	1
(A) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $xy = 10$ (C) $\exists x \in \mathbb{R}$, such that $\forall y \in \mathbb{R}, xy = 10$ (B) $\exists x \in \mathbb{R}$, such that $\forall y \in \mathbb{R}, x + y = 10$ (D) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x + y = 10$	
4. Which of the following is the Cartesian equation for a sphere with centre $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ and radius 3?	1
(A) $x^2 + x + y^2 + 2y + z^2 - 2z = 0$ (C) $x^2 + 2x + y^2 + 4y + z^2 - 4z = 0$ (B) $x^2 - x + y^2 - 2y + z^2 + 2z = 0$ (D) $x^2 - 2x + y^2 - 4y + z^2 + 4z = 0$	

5. On the argand diagram below, the twelve points $P_1, P_2, P_3, \dots, P_{12}$ are evenly spaced around the circle of radius 3. 1



Which set of points represent the solutions to $z^3 = -27i$?

- (A) P_2, P_6, P_{11} (B) P_4, P_8, P_{12} (C) P_3, P_7, P_{11} (D) P_1, P_5, P_9
6. What is the value of $\int_{-k}^k \{f(x) - f(-x)\} dx$? 1

- (A) $\int_0^k f(x) dx$ (C) 0
- (B) $4 \int_0^k f(x) dx$ (D) $2 \int_0^k f(x) dx$

7. Which expression is equal to $\int x\sqrt{1-x} dx$? 1

- (A) $-\frac{1}{3}x^2(1-x)^{\frac{3}{2}} + c$
- (B) $\frac{1}{3}x^2(1-x)^{\frac{3}{2}} + c$
- (C) $-\frac{2}{5}x(1-x)^{\frac{5}{2}} + \frac{2}{3}x(1-x)^{\frac{3}{2}} + c$
- (D) $\frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + c$

Examination continues overleaf. . .

8. Suppose $z = p + iq$ is a solution of the polynomial equation 1

$$c_4 z^4 + ic_3 z^3 + c_2 z^2 + ic_1 z + c_0 = 0$$

where p, q, c_4, c_3, c_2, c_1 and c_0 are real.

Which of the following must also be a solution?

- (A) $q + ip$ (B) $-p + iq$ (C) $-p - iq$ (D) $-p - iq$

9. The line ℓ_1 has vector equation $\tilde{r}_1 = \tilde{i} + \lambda(\tilde{j} - \tilde{k})$ and the line ℓ_2 has vector equation 1

$$\tilde{r}_2 = \left(3\tilde{i} + 2\tilde{j} - \tilde{k}\right) + \mu\left(2\tilde{i} + 2\tilde{k}\right), \text{ where } \lambda, \mu \in \mathbb{R}.$$

Which of the following statements is correct?

- (A) ℓ_1 and ℓ_2 are parallel. (C) ℓ_1 and ℓ_2 are perpendicular.
(B) ℓ_1 and ℓ_2 intersect at a point. (D) ℓ_1 and ℓ_2 are skew.

10. A ball is thrown vertically up with an initial velocity of $7\sqrt{6} \text{ ms}^{-1}$, and it is subject to gravity and air 1
resistance. The acceleration of the ball is given by $\ddot{x} = -(9.8 + 0.1 v^2) \text{ ms}^{-2}$, where x metres is its
vertical displacement from the point of projection, and $v \text{ ms}^{-1}$ is its velocity at time t seconds.

Which of the following is the time, in seconds, taken for the ball to reach its maximum height?

- (A) $\frac{10\pi}{21\sqrt{2}}$ (C) $\log_e 4$
(B) $\frac{5\pi}{21\sqrt{2}}$ (D) $\frac{\pi}{3}$

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available.
Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (13 marks) Commence a New booklet

Marks

(a) If $a + bi = \frac{2 + 4i}{1 - i}(b + i)$, where a and b are real constants, find the values of a and b . 2

(b) For real numbers $a, b > 0$ prove that $\frac{a}{b} + \frac{b}{a} \geq 2$. 2

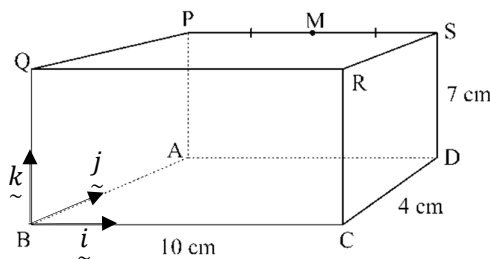
(c) The complex number z is given by $z = -p + pi$, where p is a positive real number.

It is given that $w = \frac{\sqrt{2} \bar{z}}{z^4}$.

i. Express w in the form $re^{i\theta}$, in terms of p , where $r > 0$ and $-\pi < \theta \leq \pi$. 2

ii. Find the smallest positive whole number n such that $\text{Re}(w^n) = 0$. 2

(d) The unit vectors along $\overrightarrow{BC}$, $\overrightarrow{BA}$, and $\overrightarrow{BQ}$ are $\hat{i}$, $\hat{j}$, and $\hat{k}$ respectively with each of its magnitude 1 cm. M is the midpoint of PS of the rectangular prism.



i. Find $\overrightarrow{DM}$. 1

ii. Using vector method, find the size of $\angle QDM$, giving your answer to the nearest degree. 2

(e) Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{\sin x} \cos^3 x \, dx$ 2

Examination continues overleaf. . .

Question 12 (13 marks)**Commence a New booklet****Marks**

- (a) Consider the statement:

$$\forall x \in \mathbb{R}, (x \geq 3) \Rightarrow (x^2 > 5)$$

- i. Write the contrapositive. 1
- ii. Write the negation. 1

- (b) A motorist is travelling at a constant speed of 20 ms^{-1} . When the motorist reaches a horizontal section of the road, the brakes are applied. The combined retarding force from the brakes and the friction of the road, is proportional to the speed v of the car. After travelling 80 metres along this section of the road, the speed of the car has fallen to 10 ms^{-1} .

Let x metres be the distance of the car from the start of the horizontal section.

- i. Show that $\ddot{x} = -kv$, where k is a constant. 1
- ii. Find the value of k . 2
- iii. How long did it take for the speed to drop from 20 ms^{-1} to 10 ms^{-1} ? Give the answer as an exact value. 2

- (c) Using partial fractions, show that $\int_0^1 \frac{5(1-x)}{(1+x)(3-2x)} dx = \ln \frac{4}{\sqrt{3}}$. 4

- (d) i. Sketch the graph of the set of points z defined by $|z - (3 - 4i)| = 3$, where $z \in \mathbb{C}$. 1
- ii. P is a point on the graph drawn in part (i) such that the modulus of the complex number represented by P is the smallest. 1

Find the complex number represented by P in $a + ib$ form.

Question 13 (12 marks) **Commence a New booklet** **Marks**

(a) For integers a and b , prove that if $a + b$ is odd then $a^2 + b^2$ is odd. **2**

(b) Let $f(x) = x - \ln(1 + x)$ and $g(x) = x + \ln(1 - x)$, where $0 \leq x < 1$.

i. By differentiating $f(x)$, show that $\ln(1 + x) < x$ for $0 < x < 1$. **2**

ii. By differentiating $g(x)$, show that $-\ln(1 - x) > x$ for $0 < x < 1$. **2**

iii. Deduce from (i) and (ii) that **2**

$$\ln(n + 1) - \ln n < \frac{1}{n} < \ln n - \ln(n - 1)$$

for all positive integer $n > 1$.

iv. Hence or otherwise show that $6.21 < \sum_{k=2}^{1000} \frac{1}{k} < 6.91$. **2**

(c) One of the roots of the equation $3z^3 + 13z^2 + 20z + 14 = 0$ is $-1 + i$. **2**

Find the other roots of the equation.

Examination continues overleaf...

Question 14 (17 marks) **Commence a New booklet** **Marks**

- (a) i. The line ℓ_1 has Cartesian equation $x = -y = \frac{z}{2}$. 1

Show that its vector equation is $\vec{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, where $\lambda \in \mathbb{R}$.

- ii. Write the vector equation of the line ℓ_2 in the form $\vec{r} = \vec{a} + \mu \vec{b}$ that passes 1
through the point $A(1, 1, 0)$ and is parallel to the vector $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, where $\mu \in \mathbb{R}$.

- iii. Find the acute angle θ between the lines ℓ_1 and ℓ_2 . 1

- iv. Find the coordinates of the point of intersection N of the lines ℓ_1 and ℓ_2 . 1

- v. Find the shortest distance from the point A to the line ℓ_1 . 2

- vi. Find the equation of a line ℓ_3 which bisects the acute angle θ and passes through N 2
such that the three lines ℓ_1, ℓ_2 and ℓ_3 lie on the same plane.
[You may consider the unit vectors of the directional vectors of lines ℓ_1 and ℓ_2].

- (b) Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$, for $n \in \mathbb{Z}^+$.

- i. Using integration by parts, show that $nI_n = (n-1)I_{n-2}$, where $n \geq 2$. 2

A sequence $\{x_n\}$ is defined by $x_n = nI_n I_{n-1}$ for $n \in \mathbb{Z}^+$.

- ii. Using part (i) or otherwise, show that $x_{n+1} = x_n$. 1

- iii. Hence show that $x_1 = x_2 = x_3 = \dots = x_n = x_{n+1} = \frac{\pi}{2}$. 2

- iv. Explain why $I_n \leq I_{n-1}$. 1

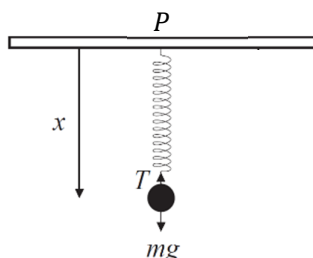
- v. Using parts (ii), (iii) and (iv) or otherwise, show $\sqrt{\frac{\pi}{2(n+1)}} \leq I_n \leq \sqrt{\frac{\pi}{2n}}$ for $n \in \mathbb{Z}^+$. 3

Question 15 (18 marks) **Commence a New booklet**

Marks

- (a) One end of a light spring of natural length l m is tied to a fixed point P at the ceiling and the other end to a particle of mass m kg as shown in the diagram below.

Assume there is no air resistance. g is the constant acceleration due to gravity.



The particle is initially pulled down to a distance of 6 m from the ceiling and from there it is projected downwards at a speed of 3.5 ms^{-1} . The particle then oscillates vertically in simple harmonic motion.

Let x be the displacement of the particle from the ceiling.

For the particle, the force T exerted by the spring is proportional to $x - l$, that is $T = k(x - l)$ newtons where k is the stiffness of the spring which is a positive constant.

- i. Show that $\ddot{x} = -\frac{k}{m} \left[x - \left(\frac{mg}{k} + l \right) \right]$. 1
- ii. Show that $x = a \cos \left(\sqrt{\frac{k}{m}} t + \alpha \right) + \frac{mg}{k} + l$ satisfies the differential equation in part (i), 1
where a and α are constants.

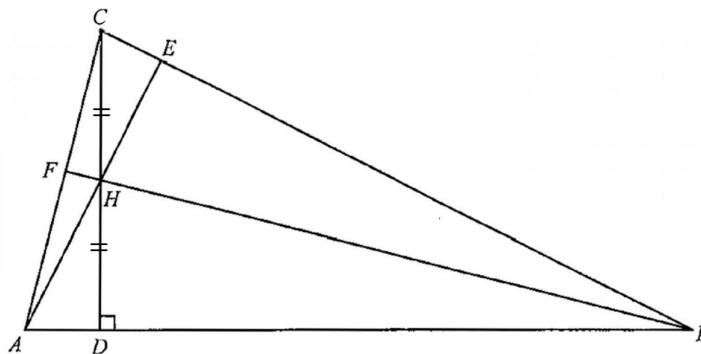
It is given that $m = 4 \text{ kg}$, $l = 5 \text{ m}$, $k = 49 \text{ Nm}^{-1}$ and $g = 9.8 \text{ ms}^{-2}$.

- iii. Find the center of motion. 1
- iv. Find the period and amplitude of the motion. 3
- v. Find the values of T at the extremes of the motion. 2

Examination continues overleaf. . .

(b)

Marks



CD is an altitude of the $\triangle ABC$ and H is a mid point of CD . AH and BH are produced to meet BC and AC at E and F respectively.

Let $\vec{p}$, $\lambda \vec{p}$ ($\lambda > 1$) and $\vec{q}$ be $\overrightarrow{AD}$, $\overrightarrow{AB}$ and $\overrightarrow{DH}$ respectively. Let $\frac{BE}{EC} = r$.

i. Find $\overrightarrow{AC}$ in terms of $\vec{p}$ and $\vec{q}$. 1

ii. Show that $\overrightarrow{AE} = \frac{(r + \lambda)\vec{p} + 2r\vec{q}}{1 + r}$. 2

iii. Using the fact that A, H and E are collinear, show that $r = \lambda$. 2

It is given that $|\vec{p}| = 1$ and $|\vec{q}| = 2$ and H is the orthocentre of the triangle $\triangle ABC$.

[The orthocenter is the point where all the three altitudes of the triangle intersect each other].

iv. Using AH is perpendicular to BC or otherwise, find the value of λ . 2

v. Using the fact that B, H and F are collinear or otherwise, find the ratio $AF:FC$ 3

Question 16 (17 marks)

Commence a New booklet

Marks

- (a) A food parcel is dropped vertically from a rescue helicopter which is 2000 metres above a group of stranded refugees in a war-torn country. After 10 seconds a parachute opens automatically. Air resistance is neglected for the first 10 seconds but then the effect of the open parachute applies a resistance of $2Mv$ newtons where M kg is the mass of the parcel plus parachute and $v \text{ ms}^{-1}$ is the velocity after t seconds ($t \geq 10$ seconds).

Take the position of the helicopter to be the origin, the downwards direction as positive and the value of g , the acceleration due to gravity, as 10 ms^{-2} .

- i. Show that the velocity of the parcel at the end of 10 seconds is 100 ms^{-1} and the distance fallen at the end of 10 seconds is 500 metres. 2

- ii. Show that the velocity of the parcel after the parachute opens is given by 2

$$v = 5 + 95e^{-2(t-10)}$$

for $t \geq 10$.

- iii. Find x , the distance fallen as a function of t . 2

- (b) Two sequences $u_1, u_2, u_3, \dots$ and $v_1, v_2, v_3, \dots$ are given by

$$u_1 = 1, \quad v_1 = 1 \quad \text{and}$$

$$u_{n+1} = u_n + 3v_n, \quad v_{n+1} = 2u_n + 7v_n$$

for positive integers n .

- i. Using Mathematical induction, prove that $2u_n^2 - 3v_n^2 + 6u_nv_n = 5$ for all positive integer n . 2

The sequence $r_1, r_2, r_3, \dots$ is such that $r_n = \frac{u_n}{v_n}$ for positive integers n .

It is given that as $n \rightarrow \infty$, $v_n \rightarrow \infty$ and $r_n \rightarrow L$ for some real constant L .

- ii. Using the result in (i) or otherwise, show that $L = \frac{1}{2}(\sqrt{15} - 3)$. 2

Examination continues overleaf...

- (c) Let $\omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$ and $S_n = \sum_{r=1}^n \omega^r$, where n is a positive integer. **Marks**

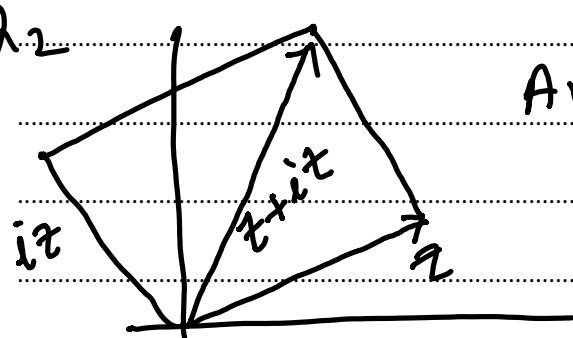
It is given that $1 + \omega + \omega^2 = 0$. It is also given that $S_n = 0$ if n is multiple of 3. (DO NOT prove this).

- i. Find S_n if n is a not multiple of 3. **2**
- ii. Prove that there exists **no** integer m such that $(S_{2022} + S_{2023} + S_{2024})^m = 2$. **2**
- iii. Find all positive integers k such that $(S_n)^k + (S_{n+1})^k + (S_{n+2})^k = 2$. **3**

End of paper

Sample solutions - ext 2 trial - 2023

Q₁ $4+3i = 5\left(\frac{4}{5} - \frac{3}{5}i\right)$
 $= 5e^{i\theta}, \tan\theta = \frac{3}{4}$
 $\approx 5e^{0.644i}$ (D)

Q₂ 
$$\text{Area} = \frac{1}{2} |z| |iz|$$
$$= \frac{1}{2} |z|^2$$
 (C)

Q₃ (D)

Q₄ $(x-1)^2 + (y-2)^2 + (z+2)^2 = 9$
 $x^2 - 2x + y^2 - 4y + z^2 + 4z = 0$
(D)

Q₅ $z^3 = -27i = 27e^{\frac{3\pi}{2}i}$
 $z = 3e^{\frac{1}{3}(2k\pi + \frac{3\pi}{2})i}; k = -2, -1, 0$
(B)

Q6

$$\begin{aligned} g(x) &= f(x) - f(-x) \\ g(-x) &= f(-x) - f(x) \\ &= -g(x) \end{aligned}$$

(C)

$$\begin{aligned} Q7 \quad \int x \sqrt{1-x} &= - \int (1-x-1)(1-x)^{1/2} dx \\ &= - \int (1-x)^{3/2} - (1-x)^{1/2} dx \\ &= \frac{2}{5}(1-x)^{5/2} - \frac{2}{3}(1-x)^{3/2} + C \end{aligned}$$

(D)

Q8

$$C_4 z^4 + i C_3 z^3 + C_2 z^2 + i C_1 z + C_0 = 0$$

$$C_4 (iz)^4 - C_3 (iz)^3 - C_2 (iz)^2 + C_1 (iz) + C_0 = 0$$

let $w = iz$

$$C_4 w^4 - C_3 w^3 - C_2 w^2 + C_1 w + C_0 = 0$$

$w = i(P+iQ)$ is a solution
 $= -Q + iP$

$iz_1 = w_1 = -Q - iP$ is also a solⁿ

$\times (-i)$

$$z_1 = -P + Qi$$

(B)

$$Q_9 \quad r_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$r_2 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

$$1 = 3 + 2\mu \Rightarrow \mu = -1$$

$$\lambda = 2$$

$$-2 = -1 + 2\mu$$

$$-2 \neq -1 - 2 \therefore \textcircled{D}$$

Q10

$$\int_{7\sqrt{6}}^0 \frac{1}{9.8 + 0.1v^2} dv = \int_0^T -dt$$

$$-T = \frac{1}{\sqrt{9.8} \cdot 0.1} \left[\tan^{-1} \left(\frac{\sqrt{0.1} v}{\sqrt{9.8}} \right) \right]_{7\sqrt{6}}^0$$

$$T = \frac{10}{\sqrt{98}} \tan^{-1} \left(\frac{\sqrt{0.1} \times 7\sqrt{6}}{\sqrt{9.8}} \right)$$

$$= \frac{10}{7\sqrt{2}} \tan^{-1}(\sqrt{3}) = \frac{10\pi}{21\sqrt{2}} \text{ sec}$$

$\textcircled{A}$

- 1 - ☐ A ☐ B ☐ C ☒ D
- 2 - ☐ A ☐ B ☒ C ☐ D
- 3 - ☐ A ☐ B ☐ C ☒ D
- 4 - ☐ A ☐ B ☐ C ☒ D
- 5 - ☐ A ☒ B ☐ C ☐ D

- 6 - ☐ A ☐ B ☒ C ☐ D
- 7 - ☐ A ☐ B ☐ C ☒ D
- 8 - ☐ A ☒ B ☐ C ☐ D
- 9 - ☐ A ☐ B ☐ C ☒ D
- 10 - ☒ A ☐ B ☐ C ☐ D

Q11

$$a) a+ib = \frac{2+4i}{1-i} (b+i) \times \frac{1+i}{1+i}$$

$$= \frac{2(1+2i)(1+i)(b+i)}{2}$$

$$= (-1+3i)(b+i)$$

$$= -b-3 + (3b-1)i$$

$$a = -b-3 \quad a=1 \quad b = 3b-1$$

$$a = -\frac{7}{2} \quad \checkmark$$

$$b = \frac{1}{2} \quad \checkmark$$

b)

$$\left(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}} \right)^2 \geq 0 \quad \checkmark$$

$$\frac{a}{b} - 2 + \frac{b}{a} \geq 0$$

$$\frac{a}{b} + \frac{b}{a} \geq 2 \quad \checkmark$$

$$c) i) z = -p+pi = p(-1+i)$$

$$= \sqrt{2}p e^{i\left(\frac{3\pi}{4}\right)} \quad \checkmark$$

$$w = \frac{\sqrt{2} \bar{z}}{z^4} = \frac{1 p e^{-i\left(\frac{3\pi}{4}\right)}}{4 p^4 e^{i\left(3\pi\right)}}$$

ii)



$$n = 8k \pm 2$$



d).





.....

✓



$Q = 26.37098991 \div 26^0$ ✓

$$\begin{aligned}
 e) \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \cos^3 x \, dx &= \int_0^{\frac{\pi}{2}} \sqrt{\sin x} (1 - \sin^2 x) \, d(\sin x) \\
 &= \left[\frac{2}{3} (\sin x)^{3/2} - \frac{2}{7} (\sin x)^{7/2} \right]_0^{\frac{\pi}{2}} \\
 &= \frac{2}{3} - \frac{2}{7} = \frac{8}{21}
 \end{aligned}$$

Q2

a) i) $\forall x \in \mathbb{R} \quad (x^2 \leq 5) \Rightarrow (x < 3)$ ✓

ii) $\exists x \in \mathbb{R}$ such that $(x \geq 5) \wedge (x^2 \leq 5)$ ✓

b)

$k, v \rightarrow v$
 $\leftarrow \text{eq}$

i)

$f = ma \rightarrow m \dot{x} = -k_1 v$

$\dot{x} = -\frac{k_1}{m} v$ ✓

$\dot{x} = -k v$ where $k = \frac{k_1}{m}$

ii)

$v \frac{dv}{dx} = -k v$

$\int_{20}^{10} 1 dv = \int_0^{\infty} -k dx$

$10 - 20 = -k(80 - 0)$ ✓

$-10 = -80k$

$k = \frac{1}{8}$ ✓

iii)

$$10 \frac{dv}{dt} = -\frac{1}{8} v$$

$$\int_{20}^{10} \frac{1}{v} dv = \int_0^T -\frac{1}{8} dt \quad \checkmark$$

$$\ln 10 - \ln 20 = -\frac{T}{8} \quad \checkmark$$

$$T = 8 \ln 2 \text{ sec}$$

c)

$$\frac{5(1-x)}{(1+x)(3-2x)} = \frac{5(x-1)}{(x+1)(2x-3)} = \frac{A}{x+1} + \frac{B}{2x-3}$$

$$5(x-1) = A(2x-3) + B(x+1)$$

$$x = -1$$

$$-10 = -5A \Rightarrow A = 2 \quad \checkmark$$

$$x = \frac{3}{2}$$

$$\frac{5}{2} = B\left(\frac{5}{2}\right) \Rightarrow B = 1 \quad \checkmark$$

$$\therefore \frac{5(1-x)}{(1+x)(3-2x)} = \frac{2}{x+1} + \frac{1}{2x-3}$$

OR

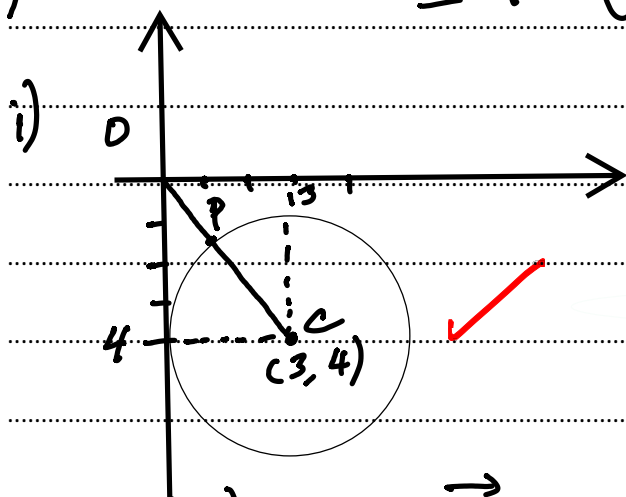
$$\frac{5(x-1)}{(x+1)(2x-3)} = \frac{5(x+1-2)}{(x+1)(2x-3)} = \frac{5}{2x-3} - \frac{10}{(x+1)(2x-3)} \quad \checkmark$$

$$= \frac{5}{2x-3} + 2 \left[\frac{1}{x+1} - \frac{2}{2x-3} \right]$$

$$= \frac{1}{2x-3} + \frac{2}{x+1} \quad \checkmark$$

$$\begin{aligned}
 \int_0^1 \frac{5(1-x)}{(1+x)(3-2x)} dx &= \int_0^1 \frac{1}{2x-3} + \frac{2}{x+1} dx \\
 &= \left[\frac{1}{2} \ln |2x-3| + 2 \ln |x+1| \right]_0^1 \\
 &= 0 + 2 \ln 2 - \left(\frac{1}{2} \ln 3 \right) \\
 &= \ln 4 - \ln \sqrt{3} \\
 &= \ln \left(\frac{4}{\sqrt{3}} \right)
 \end{aligned}$$

d)



ii) $\vec{OP} = \frac{2}{5} \vec{OC}$

$\therefore$ the complex number represented by P

$$\begin{aligned}
 &= \frac{2}{5} (3-4i) \\
 &= \frac{6}{5} - \frac{8}{5}i
 \end{aligned}$$

Q.13

a) If both a and b odd or both even
then $a+b$ is even

$\therefore$ One of them is odd and other is even

w.l.g take a is odd and b is even

then $\exists m, n \in \mathbb{Z}$ such that

$$a = 2m+1 \text{ and } b = 2n$$

$$a^2 + b^2 = (2m+1)^2 + 4n^2$$

$$= 2(2m^2 + 2n^2 + 2m) + 1$$

$$= 2k+1 \text{ where } k = 2m^2 + 2n^2 + 2m \in \mathbb{Z}$$

$\therefore a^2 + b^2$ is odd

b)

i) $f(x) = x - \ln(1+x)$

$$f'(x) = 1 - \frac{1}{1+x} = \frac{x}{x+1} > 0 \text{ for } x \in (0,1)$$

$$\text{and } f(0) = f'(0) = 0$$

$$\therefore f(0) < f(x) \quad \forall x \in (0,1)$$

$$0 < x - \ln(1+x)$$

$$\therefore \ln(1+x) < x \quad \forall x \in (0,1)$$

— (1)

$$ii) \quad g(x) = x + \ln(1-x)$$

$$g'(x) = 1 - \frac{1}{1-x} = \frac{-x}{1-x}$$

$$= \frac{x}{x-1} < 0 \quad \forall x \in (0,1)$$

$$\text{and } g(0) = 0$$

$$\therefore g(0) > g(x) \quad \forall x \in (0,1)$$

$$0 > x + \ln(1-x)$$

$$\therefore -\ln(1-x) > x \quad \forall x \in (0,1) \quad \text{--- (2)}$$

iii)

From (1) & (2)

$$\ln(1+x) < x < -\ln(1-x)$$

$$\text{Sub } x = \frac{1}{n} \text{ for } n > 1$$

$$\ln\left(1 + \frac{1}{n}\right) < \frac{1}{n} < -\ln\left(1 - \frac{1}{n}\right)$$

$$\ln(1+n) - \ln n < \frac{1}{n} < \ln n - \ln(n-1)$$

iv)

$$n=2 \quad \ln 3 - \ln 2 < \frac{1}{2} < \ln 2 - \ln 1$$

$$n=3 \quad \ln 4 - \ln 3 < \frac{1}{3} < \ln 3 - \ln 2$$

$\vdots$

$$n=1000 \quad \ln 1001 - \ln 1000 < \frac{1}{1000} < \ln 1000 - \ln 999$$

+

$$\ln 1001 - \ln 2 < \sum_{k=2}^{1000} \frac{1}{k} < \ln 1000$$

$$6.21560.. = \ln \frac{1001}{2} < \sum_{k=2}^{1000} \frac{1}{k} < \ln 1000 = 6.9077..$$

$$6.21 < \sum_{k=2}^{1000} \frac{1}{k} < 6.91$$

c)

$$3z^3 + 13z^2 + 20z + 14 = 0$$

$$\text{roots: } -1+i, -1-i, \alpha$$

$$-2 + \alpha = -\frac{13}{3}$$

$$\alpha = -\frac{7}{3}$$

Q14

a) i) $x = -y = \frac{z}{2} = a$

$l_1: \underline{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ -a \\ 2a \end{pmatrix} = a \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ ✓

ii) $l_2: \underline{r} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ ✓

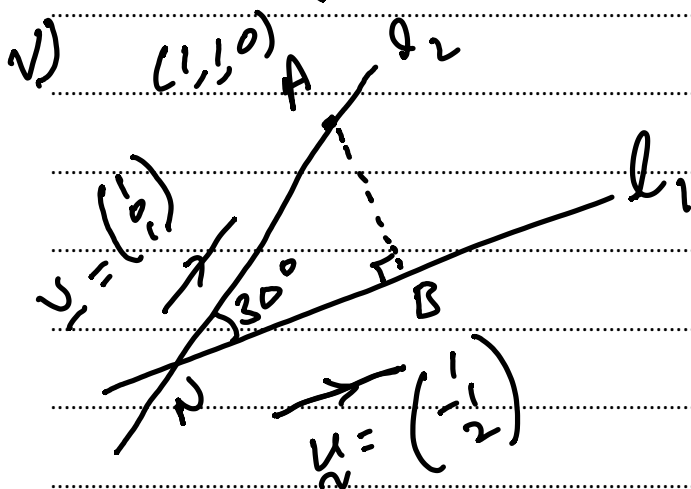
iii) $\cos Q = \frac{1+0+2}{\sqrt{6} \sqrt{2}} = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$

$Q = 30^\circ$ ✓

iv) $a \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1+\mu \\ 1 \\ \mu \end{pmatrix} \Rightarrow \begin{matrix} a = 1+\mu \\ a = -1 \\ \mu = -2 \end{matrix}$

check $2a = \mu$ when $a = -1, \mu = -2$

$\therefore N(-1, 1, -2)$ ✓



method 1

$AN = \sqrt{2^2 + 0^2 + 2^2} = 2\sqrt{2}$ ✓

$AB = 2\sqrt{2} \sin 30^\circ = \sqrt{2}$ units ✓

method 2 $\vec{NA} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$

$$\vec{NB} = \text{proj}_{\vec{u}} \vec{NA}$$

$$= \frac{\vec{NA} \cdot \vec{u}}{|\vec{u}|^2} \vec{u}$$

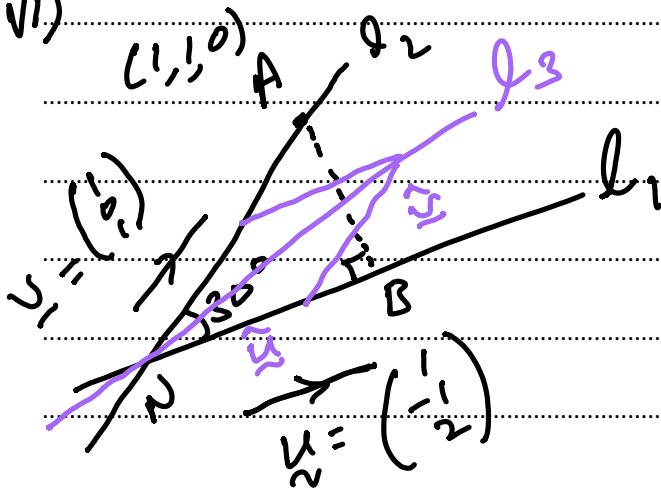
$$= \frac{2+6}{8} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad \checkmark$$

$$\vec{AB} = \vec{AN} + \vec{NB}$$

$$= \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$$

$$|\vec{AB}| = \sqrt{2} \quad \checkmark$$

vi)



Direction Vector of l_3

$$= \vec{u} + \vec{v}$$

$$= \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \checkmark$$

$$= \frac{1}{\sqrt{6}} \begin{pmatrix} 1+\sqrt{3} \\ 2 \\ 2+\sqrt{3} \end{pmatrix}$$

Equation of l_3

$$\vec{r} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1+\sqrt{3} \\ 2 \\ 2+\sqrt{3} \end{pmatrix}; \lambda \in \mathbb{R} \quad \checkmark$$

$$\begin{aligned}
 I_n &= \int_0^{\frac{\pi}{2}} \sin^n x \, dx = \int_0^{\frac{\pi}{2}} \sin x \sin^{n-1} x \, dx \\
 &= \left[\sin^n x (-\cos x) \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx \\
 &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) \, dx
 \end{aligned}$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$I_n (1 + n-1) = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$x_n = n I_n I_{n-1}$$

$$= (n-1) I_{n-2} I_{n-1}$$

$$= (n-1) I_{n-1} I_{n-2} = x_{n-1}$$

$$x_1 = 1 I_1 I_0$$

$$I_0 = \int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2}$$

$$I_1 = \int_0^{\frac{\pi}{2}} \sin x \, dx = [-\cos x]_0^{\frac{\pi}{2}} = 1$$

$$x_1 = \frac{\pi}{2}$$

but by ii $x_n = x_{n-1} = x_{n-2} = \dots = x_2 = x_1$ ✓
 $= \frac{\pi}{2}$

✓ $0 \leq \sin x \leq 1 \quad \forall x \in [0, \frac{\pi}{2}]$
 $\therefore \sin^n x \leq \sin^{n-1} x \quad \forall x \in [0, \frac{\pi}{2}]$

$\therefore I_n \leq I_{n-1}$

$\frac{\pi}{2} = x_n = n I_n I_{n-1}$ by ii
 $\therefore \frac{\pi}{2} \geq n I_n I_n$ by iii
 $I_n^2 \leq \frac{\pi}{2n}$ ✓

$I_n \leq \sqrt{\frac{\pi}{2n}} \quad \text{--- (*)}$

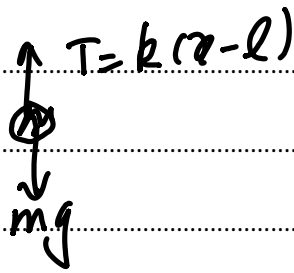
$\frac{\pi}{2} = x_{n+1} = (n+1) I_{n+1} I_n$
 $\leq (n+1) I_n I_n$ ✓
 $I_n^2 \geq \frac{\pi}{2(n+1)}$

$I_n \geq \sqrt{\frac{\pi}{2(n+1)}} \quad \text{--- (**)}$

from (*), (**)
 $\sqrt{\frac{\pi}{2(n+1)}} \leq I_n \leq \sqrt{\frac{\pi}{2n}}$ ✓

Q15

a)



i) $F = ma \downarrow \quad m\ddot{x} = mg - k(x - l)$

$$\ddot{x} = g - \frac{k}{m}(x - l)$$

$$= -\frac{k}{m} \left(x - l - \frac{mg}{k} \right) \checkmark$$

$$= -\frac{k}{m} \left(x - \left(\frac{mg}{k} + l \right) \right)$$

ii)

$$x = a \cos \left(\sqrt{\frac{k}{m}} t + \alpha \right) + \frac{mg}{k} + l$$

$$\dot{x} = -a \sqrt{\frac{k}{m}} \sin \left(\sqrt{\frac{k}{m}} t + \alpha \right)$$

$$\ddot{x} = -\frac{ak}{m} \cos \left(\sqrt{\frac{k}{m}} t + \alpha \right) \checkmark$$

$$= -\frac{k}{m} \left(x - \left(\frac{mg}{k} + l \right) \right)$$

$\therefore x = a \cos \left(\sqrt{\frac{k}{m}} t + \alpha \right) + \frac{mg}{k} + l$ satisfies the DE.

$$\text{iii)} \quad \frac{mg}{k} + l = \frac{4 \times 9.8}{49} + 5 = 5.8$$

$$\ddot{x} = -\frac{49}{4} (x - 5.8)$$

$\therefore$ Centre of motion at $x = 5.8 \text{ m}$.

$$\text{iv)} \quad \text{period} = P = \frac{2\pi}{\sqrt{\frac{49}{4}}} = \frac{4\pi}{7} \text{ sec}$$

method 1

$$t = 0 \quad x = 6 \text{ m}, \quad \dot{x} = 3.5 \text{ m s}^{-1}$$

$$6 = a \cos \alpha + 5.8$$

$$0.2 = a \cos \alpha \quad \text{--- (1)}$$

$$3.5 = -a \sqrt{\frac{49}{4}} \sin(\alpha)$$

$$1 = -a \sin \alpha \quad \text{--- (2)}$$

$$\text{(1)}^2 + \text{(2)}^2$$

$$0.2^2 + 1^2 = a^2$$

$$a = \sqrt{1.04} = 1.01980 \dots$$

$$\approx 1.02 \text{ m}$$

method 2

$$v^2 = n^2 (a^2 - (x - c)^2)$$

$$v^2 = \frac{49}{4} (a^2 - (x - 5.8)^2)$$

$$\text{sub } x = 6 \quad v = 3.5$$

$$3.5^2 = \frac{49}{7} (a^2 - (6-5.8)^2)$$

$$1 = a^2 - (0.2)^2 \Rightarrow a^2 = 1.04$$

$$a = \sqrt{1.04}$$

Method 3

$$\ddot{x} = -\frac{49}{4} (x - 5.8)$$

$$v \frac{dv}{dx} = -\frac{49}{4} (x - 5.8)$$

$$\int_{3.5}^0 v dv = -\frac{49}{4} \int_6^{x_1} (x - 5.8) dx$$

$$\left[\frac{v^2}{2} \right]_{3.5}^0 = -\frac{49}{4} \left[\frac{x^2}{2} - 5.8x \right]_6^{x_1}$$

$$-\frac{3.5^2}{2} = -\frac{49}{4} \left[\frac{x_1^2}{2} - 5.8x_1 - \left(\frac{36}{2} - 5.8 \times 6 \right) \right]$$

$$\frac{1}{2} = \frac{x_1^2}{2} - 5.8x_1 + 16.8 \Rightarrow x_1^2 - 11.6x_1 + 32 - 6 = 0$$

$$x_1 = 6.81982, 4.7802$$

$$\therefore a = 6.81982 - 5.8 = 1.0198 \div 1.02$$

v)

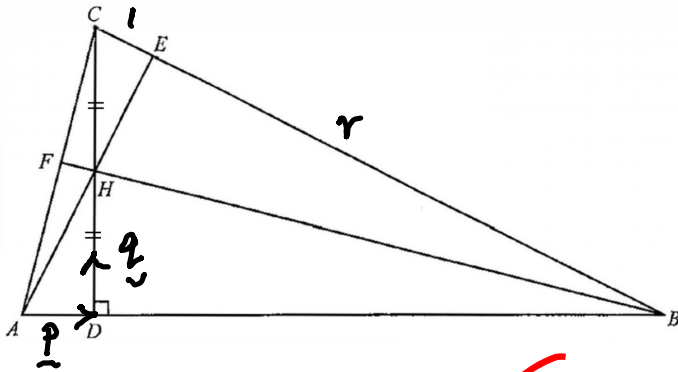
Extremes are at $x = 5.8 \pm \sqrt{1.04}$ m

$$\text{At } x = 5.8 + \sqrt{1.04} \quad T = 49(5.8 + \sqrt{1.04} - 5)$$

$$= 89.17 \text{ N}$$

$$\text{At } x = 5.8 - \sqrt{1.04} \quad T = -10.77 \text{ N}$$

b)



$$\vec{AB} = \lambda \vec{p}$$

$$\vec{BE} = r \vec{EC}$$

$$i) \vec{AC} = \vec{p} + 2\vec{q}$$

$$ii) \vec{AE} = \vec{AC} + \vec{CE}$$

$$= \vec{AC} + \frac{1}{1+r} \vec{CB}$$

$$= \vec{AC} + \frac{1}{1+r} (\vec{CA} + \vec{AB})$$

$$= \left(1 - \frac{1}{1+r}\right) \vec{AC} + \frac{1}{1+r} \vec{AB}$$

$$= \frac{r}{r+1} (\vec{p} + 2\vec{q}) + \frac{1}{1+r} (2\vec{p})$$

$$= \frac{(r+2)\vec{p} + 2r\vec{q}}{1+r}$$

iii)

Since A, H and E are collinear

$$\vec{AH} = \mu \vec{AE}$$

$$\vec{p} + \vec{q} = \frac{\mu}{1+r} [(r+2)\vec{p} + 2r\vec{q}]$$

$$\therefore 1 = \frac{\mu(r+2)}{1+r} \text{ and } 1 = \frac{2r\mu}{1+r}$$

$$\therefore \mu(r+d) = 2r \mu$$

$$d+r = 2r$$

$$d = r \quad \checkmark$$

iv)

$$\vec{AH} \cdot \vec{BC} = 0$$

$$\vec{AH} \cdot (\vec{BA} + \vec{AC}) = 0$$

$$(\underline{p} + \underline{q}) \cdot (-d \underline{p} + \underline{p} + 2\underline{q}) = 0 \quad \checkmark$$

$$(\underline{p} + \underline{q}) \cdot ((1-d)\underline{p} + 2\underline{q}) = 0$$

$$(1-d)|\underline{p}|^2 + (2+1-d)\underline{p} \cdot \underline{q} + 2|\underline{q}|^2 = 0$$

$$1-d + (3-d)0 + 2(4) = 0$$

$$d = 9 \quad \checkmark$$

v)

$$\text{Let } \vec{AF} = m \vec{AC} \text{ and } \vec{BH} = r \vec{BF}$$

$$\vec{AF} = m(\underline{p} + 2\underline{q}) \quad \checkmark$$

$$\vec{AF} = \vec{AB} + \vec{BF}$$

$$= \vec{AB} + \frac{1}{r} \vec{BH}$$

$$= \vec{AB} + \frac{1}{r} (\vec{BD} + \vec{DH})$$

$$= 9\underline{p} + \frac{1}{r} (-8\underline{p} + \underline{q})$$

$$= (9 - \frac{8}{r})\underline{p} + \frac{1}{r}\underline{q} \quad \checkmark$$

$$\therefore m\vec{p} + 2m\vec{q} = \left(9 - \frac{8}{\gamma}\right)\vec{p} + \frac{1}{\gamma}\vec{q}$$

$$m = 9 - \frac{8}{\gamma} \quad \text{and} \quad 2m = \frac{1}{\gamma}$$

$$m = 9 - 8(2m)$$

$$17m = 9$$

$$m = \frac{9}{17}$$

$$\vec{AF} = \frac{9}{17} \vec{AC} \quad \therefore AF:FC = 9:8$$

Method 2 Let $\vec{AF} = m\vec{AC} = m(\vec{p} + 2\vec{q})$

$$\vec{AC} \cdot \vec{FH} = 0 \Rightarrow \vec{AC} \cdot (\vec{AH} - \vec{AF}) = 0$$

$$(\vec{p} + 2\vec{q}) \cdot (\vec{p} + \vec{q} - m(\vec{p} + 2\vec{q})) = 0$$

$$(\vec{p} + 2\vec{q}) \cdot ((1-m)\vec{p} + (1-2m)\vec{q}) = 0$$

$$(1-m)|\vec{p}|^2 + (1-2m)\vec{p} \cdot \vec{q} + 2(1-m)\vec{p} \cdot \vec{q} + 2(1-2m)|\vec{q}|^2 = 0$$

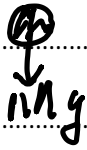
$$(1-m)1 + 2(1-2m) \cdot 4 = 0$$

$$1-m+8-16m=0 \Rightarrow m = \frac{9}{17}$$

$$\therefore AF:FC = 9:8$$

Q16

a) i)



$$F = ma \quad \checkmark$$

$$m\ddot{x} = mg$$

$$\ddot{x} = \frac{dv}{dt} = g = 10$$

$$\int_0^v 1 dv = \int_0^t 10 dt$$

$$v = 10t$$

$$t = 10 \quad v = 100 \text{ m s}^{-1}$$

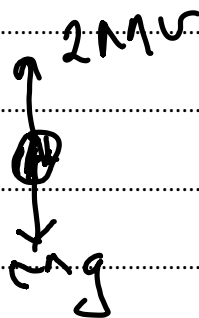
$$\frac{dx}{dt} = 10t$$

$$\int_0^x dx = \int_0^{10} 10t dt$$

$$x = \left[5t^2 \right]_0^{10} = 500 \text{ m}$$

ii)

For $t \geq 10$



$$F = ma \quad \checkmark$$

$$M\ddot{x} = Mg - 2Mv$$

$$\ddot{x} = g - 2v$$

$$= 10 - 2v$$

$$= -2(v - 5) \quad \checkmark$$

$$5 \leq v \leq 100$$

$$\int_{100}^v \frac{1}{v-5} dv = \int_{10}^t -2 dt$$

$$\left[\ln(v-5) \right]_{100}^v = -2(t-10)$$

$$\ln(v-5) - \ln(95) = -2(t-10) \quad \checkmark$$

Since $v-5 > 0$

$$\ln\left(\frac{v-5}{95}\right) = -2(t-10)$$

$$v = 5 + 95 e^{-2(t-10)}$$

$$\text{iv) } \frac{dx}{dt} = 5 + 95 e^{-2(t-10)}$$

$$\int_{500}^x dx = \left[5t - \frac{95}{2} e^{-2(t-10)} \right]_{10}^t \quad \checkmark$$

$$x - 500 = 5t - \frac{95}{2} e^{-2(t-10)} - \left(50 - \frac{95}{2} \right)$$

$$x = 5t - \frac{95}{2} e^{-2(t-10)} + \frac{99.5}{2} \quad \checkmark$$

$$S(n): 2u_n^2 - 3v_n^2 + 6u_nv_n = 5$$

prove for $S(1)$

$$L.H.S = 2u_1^2 - 3v_1^2 + 6u_1v_1$$

$$= 2 - 3 + 6$$

$$= 5 = R.H.S$$

$\therefore S(1)$ is true ✓

Assume $S(k)$ is true

$$i.e., 2u_k^2 - 3v_k^2 + 6u_kv_k = 5$$

prove for $S(k+1)$

$$L.H.S = 2u_{k+1}^2 - 3v_{k+1}^2 + 6u_{k+1}v_{k+1}$$

$$= 2(u_k + 3v_k)^2 - 3(2u_k + 7v_k)^2 + 6(u_k + 3v_k)(2u_k + 7v_k)$$

$$= 2u_k^2 + 12u_kv_k + 18v_k^2$$

$$- 12u_k^2 - 84u_kv_k - 147v_k^2$$

$$+ 12u_k^2 + 78u_kv_k + 126v_k^2$$

$$= 2u_k^2 + 6u_kv_k - 3v_k^2 \quad \checkmark$$

$= 5$ by assumption.

$= R.H.S \quad \therefore S(k+1)$ is true

$\therefore$ By MI, $S(n)$ is true for all $n \in \mathbb{N}$

$$r_n = \frac{u_n}{v_n}, \quad \lim_{n \rightarrow \infty} v_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} r_n = L$$

$$\div v_n^2 \quad 2u_n^2 - 3v_n^2 + 6u_nv_n = 5$$

$$2r_n^2 - 3 + 6r_n = \frac{5}{v_n}$$

take limit as $n \rightarrow \infty$ both sides

$$2L^2 - 3 + 6L = \lim_{n \rightarrow \infty} \frac{5}{v_n} = 0 \quad \checkmark$$

$$2L^2 + 6L - 3 = 0$$

$$L = \frac{-6 \pm \sqrt{36 + 4 \times 2 \times 3}}{4}$$

$$= \frac{-3 \pm \sqrt{15}}{2}$$

Since $u_n, v_n > 0$, $r_n > 0 \quad \forall n \quad \therefore L > 0$

$$\therefore L = \frac{1}{2}(\sqrt{15} - 3) \quad \checkmark$$

c)

$$i) \omega = e^{i\frac{2\pi}{3}} ; S_n = \sum_{r=1}^n \omega^k$$

$$S_n = \omega + \omega^2 + \dots + \omega^n \text{ and } \omega^3 = 1$$

$$= \frac{\omega(\omega^n - 1)}{\omega - 1}$$

if n is not multiple of 3 then

$$n = 3m+1 \text{ OR } n = 3m+2 \text{ for some } m \in \mathbb{Z}$$

If $n = 3m+1$

$$S_n = \frac{\omega(\omega^{3m} \times \omega - 1)}{\omega - 1} = \frac{\omega(\omega - 1)}{\omega - 1}$$

$$= \omega$$

$$\text{If } n = 3m+2 \quad S_n = \frac{\omega(\omega^{3m} \omega^2 - 1)}{\omega - 1}$$

$$= \omega(\omega + 1)$$

$$= \omega + \omega^2$$

$$= -1$$

Assume there exist an integer m

$$\text{such that } (S_{2022} + S_{2023} + S_{2024})^m = 2$$

$$2022 = 3 \times 674 \Rightarrow S_{2022} = 0$$

$$2023 = 3 \times 674 + 1 \Rightarrow S_{2023} = \omega$$

$$2024 = 3 \times 674 + 2 \Rightarrow S_{2024} = -1$$

$$\therefore (0 - 1 + \omega)^m = 2$$

$$\left(-1 - \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^m = 2$$

$$\left(-\frac{3}{2} + \frac{\sqrt{3}}{2}i\right)^m = 2$$

$$\left|-\frac{3}{2} + \frac{\sqrt{3}}{2}i\right| = \sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{3}$$

$$\sqrt{3}^m = 2$$

$$3^{\frac{m}{2}} = 2^2$$

~~Since~~ L.H.S is multiple of 3 and R.H.S = 4

$$\therefore \nexists m \in \mathbb{Z} \text{ such } (S_{2023} \pm S_{2024})^m = 2$$

$$(S_n)^k + (S_{n+1})^k + (S_{n+2})^k = 2$$

$$\{S_n, S_{n+1}, S_{n+2}\} = \{0, -1, \omega\}$$

$$\therefore 0^k + (-1)^k + \omega^k = 2$$

If k is odd $\omega^k = 3$

$$\Rightarrow |\omega|^k = 3 \Rightarrow 1 = 3 \quad \nexists$$

If k is even $\omega^k = 1$

$$\cos \frac{2k\pi}{3} = 1 = \cos 0$$

$$\frac{2k\pi}{3} = 2m\pi \text{ for } m = 0, \pm 1, \pm 2, \dots$$

$$k = 3m$$

$\therefore k$ is even and multiple of 3

$$\therefore k = 6n \text{ for } n = 0, \pm 1, \pm 2, \dots$$